CASE STUDY REVIEW REPORT
Machine Learning: A Game Changer for Additive Manufacturing Quality Assurance

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ABSTRACT

Additive Manufacturing though relatively new as compared to other production techniques, consists of a few defects along the terms of a product's structural integrity, compactness and molecular build-up from a ground level. Not only does a 3D printed product goes through the above mentioned flaws but also, has to experience issues such as High Porosity, Low Density, Residual Stresses, Stair-Stepping and Lack of Quality Consistency to name a few. This hampers the quality metrics that a product has to pass through and in turn leads to quality flaws such as scars on top surface; weak infill; overhangs or sagging sections in the print and much more. In this case study, we get to see the monitoring of manufacturing builds used during training and producing in-process quality metrics(IPQMs) for use as features of Machine Learning models. The use of these models benefits the additive manufacturing systems by tuning the builds and their training parameters with a brief description of the process. After which we observe the transition of the system from training process to QA process upon thorough monitoring of procreating builds and anomaly detection by physical analysis of the parts. This ML-QA process evaluates build quality & visualization of predictions to identify location of potential anomalies & predict real defects.

Introduction

Additive Manufacturing is a layer-based automated fabrication process for making scaled 3-dimensional physical objects directly from 3D CAD data without using part-depending tools. The main focus in terms of technical realization of additive manufacturing is solely on layers and therefore it is called "layer-based technology", "layer-oriented technology" or even "layered technology". Each layer is contoured according to the corresponding 3D data set and put on the top of the preceding one.

As Additive Manufacturing(AM), has proven to consume more time & resources in developing various products as compared to Conventional Manufacturing(CM). Even so, if the following graph is considered, a break-even point between the Cost & Volume of the units is observed. This point exists simply because of the technological advancements taking place in AM processes. When only a limited batch of unique products is to be manufactured, AM is far more competitive as compared to CM, since the cost of making a die for an instance is far more expensive than the AM options.

AM processes being relatively new in the production domain, many consider it a disruptive technology that will revolutionize production and displace automation equipment. While others have labelled this hypothesis as nothing but a "futuristic hype". But, presently what has transformed, is an outcome in which AM is complimenting current modes of manufacturing, becoming integrated with them in the factory of the future. In other words, all present indications are that AM will not displace automation technologies but instead benefit from them. Moreover, the advancing evolution of 3D printers, new market opportunities has opened for manufacturers of motion control components, lasers, 3D scanners, machine vision inspection, process control systems, robotic arms and gantry robots. All these elements of Additive Manufacturing technology need the already well-established Automation Machineries and Processes, in reaching the newer market and vice-e-versa.

This interdependency can be easily understood by taking a few examples such as, advancements in the capabilities of motion control components would give rise to more sophisticated 3D printers. Use of robotic arms could enable 3D printers to overcome size constraints, increasing not just the dimensions but also the types of objects that can be "printed". Additionally, 3D scanners manufactured by machine vision companies are already enabling 3D printers to replicate objects. Thus, Additive Manufacturing can play a vital role in the context of ACIM capitalizing on following factors such as: Design Freedom & Customization; Rapid Prototyping; Supply Chain Optimization; Reduced Material Waste; Digital Integration & Industry 4.0 and Complex Component Manufacturing. [AAA]

[TML]Looking at the limitations hindering the proliferation of AM in the current manufacturing operation including excessive time consumption, lack of real-time control, potential deviation & the transition from mass production to mass real-time customization can be helped along with Artificial Intelligent algorithms and Machine Learning(ML) models. Machine Learning for example, opens the possibility for machines to learn and improve autonomously with little or no human intervention at all. Improved decision making and efficient automation system can be achieved by applying ML models in manufacturing by deriving useful information from existing data sets, which provides a basis for approximations or predictions to operate machines. Also, it is beneficial to detect certain patterns or explore regularities in a dynamic manufacturing environment. [TML]ML for Quality Assurance(QA) metrics in AM contributes on a largely varied scale. To name a few: -

Automated Testing – improving test coverage & efficiency;

Defect Prediction & Prevention – predicting potential defects for their reduction;

Test Case Prioritization – optimizing testing resources;

Anomaly Detection – enhancing product quality by detecting anomalies or unexpected behavior;

NLP for Requirements analysis;

Performance Testing Optimization – identification of bottlenecks & similar areas.

Objectives

This Case-Study Report is based on the Case-Study of Sigma Labs' approach in acquiring & developing knowledge from applying Machine Learning models using PrintRite3D IPQM. Accordingly, few features that the paper covers are noted below as: -

i. Training ML models to recognize four common anomaly defects.
ii. Accurately predict each specific anomaly's presence with IPQM.

iii. Discuss future work required for making these techniques useful in volume production.
iv. Interpreting anomaly morphologies and their relationship with in-process data.
v. Specifying importance of using in-process measurements and non-destructive testing to generate information.
vi. Driving timely decisions on part and process quality from generated information.
vii. Developing advanced computer algorithm to accurately predict where anomalies are likely to occur.
viii. Gaining an insight towards compellingly useful characteristics of the Laser Powder Bed Fusion.
ix. Avoiding conventional quality inspections of manufactured parts that rely primarily on lengthy & expensive destructive testing.
x. Comprehending PrintRite3D metrics which includes TED, TED Sigma, TEP & TEP Sigma as features. (Beckett, n.d.)

- The current era of Industry 4.0 and upcoming Industry 5.0 is focused more towards decentralized & fully automated concept of manufacturing. This has proven to provide customers a more efficient product in an optimized manner.
- AM system manufactures product from its raw till finished form without any human intervention abiding by every norm set by these smart factories. Due to ever increasing competition and depleting natural resources, advancements in AM systems is bound to happen for the technology's sustenance. Using ML models researchers & engineers are continuously improving the Quality metrics for the desired product.
- In essence, ML in QA brings efficiency, precision and adaptability to the testing process contributing to the development of high-quality software and systems. *(Kaleem & Khan, 2020)*

LITERATURE REVIEW

A. LITERATURE SEARCH STRATEGY

The search was carried out from the assigned date till 10th of November. The three databases IEEE Explore, ASME, EBSCO Host and Research Gate were considered. Since the paper title was established from a Case-Study, it was bifurcated in three parts, viz., Additive Manufacturing (LPBF), Machine Learning (Random Forest) and Quality Assurance (PrintRite3D IPQM). Correspondingly, the search term combination: title or abstract or keywords containing these technical parameters were considered. The search resulted in __ articles that met the inclusion criteria. Furthermore, the search provided extensive results, which were then filtered out to display only journals, reports and published conference papers.

B. SELECTION CRITERIA FOR STUDY

Parsing through the resultants, only correlating concepts to the discerned title were considered for writing the report. Also, the papers which categorized under international peer-reviewed publications were the only ones taken in consideration, to guarantee high quality standards. After reviewing 18 articles, only __ articles were selected as relevant to the topic as they deal with ML models to improve QA of AM processes. Due to the segregation of the title in sub-titles and then correlation of those papers with each other on the basis of the whole title, increased the number of research papers with respect to the selection criteria. The papers with similar concept but varying in terms of applications were excluded.

C. **ASSESSING QUALITY OF PAPERS**
The quality of papers is assessed using different criteria's considered to ensure rigor, relevance & contribution to the field. The Abstract provided in each research paper was usually proven to be the detrimental factor, followed by the form of objective addressable. The 'Literature Review' presented in those papers were scrutinized for gaps in the current literature that this case-study aimed in addressing. Lastly, the contribution of new insights and experiments were examined on how they would advance the knowledge in the field. Also a neat appropriation of citations & references was checked, so as to understand the author's perception for the proposed experimentation. (Breitenbach et al., 2022)

Bibliometric Analysis: -

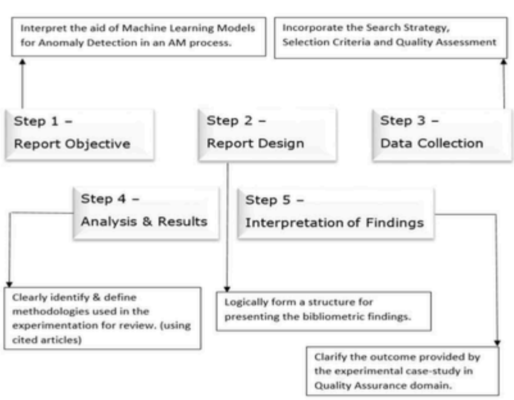

Figure 1. Research Protocol

METHODOLOGY

Laser Powder Bed Fusion (LPBF)

For the requisite case-study, Laser Powder Bed Fusion type of Additive Manufacturing process is taken into consideration for incorporating Machine Learning model. This model is then analyzed for detection of quality assurance anomalies found in the LPBF processed end-product.

Accordingly, metal AM processes are categorized into two principle techniques, namely, Powder Bed Fusion(PBF) and Directed Energy Deposition(DED). In the L-PBF process, a heat source such as a laser selectively melts the layer of powder already spread on the substrate. This process is also known as Selective Laser Melting(SLM), Directed Metal Laser Melting(DMLM), or Direct Metal Laser Sintering(DMLS). The process heats and melts the layer of powder according to the CAD file and then the material is solidified. This is done by selectively scanning specific points on the powder bed along predefined paths with the laser source energy. After finishing first layer, powder dispenser moves upwards supplying succeeding powder layer for creating new product layer. The building platform on the other hand lowers a distance equal to the layer thickness, and the recoated transfers material to the building platform, forming a new stratum. This same cycle is repeated until final shape is achieved. The un-melted powder is usually recycled post-manufacturing for several iterations, which make the L-PBF process more sustainable. (Taghian et al., 2023)

In the Case-Study, the powder selected for producing a sample part is the Ti6Al4V powder, where it is found that the major concern for recyclability of the powder is the oxygen content. Ti6Al4V is highly oxidizing at elevated temperature, this dissolved oxygen can act as a strong α-phase stabilizer that improves part strength by solid solution strengthening mechanism. Among other properties the powder has excellent strength-to-weight ratio and is highly corrosion resistant. The powder can also withstand high temperatures and is known for its low density making it significantly light-weight, along with having good fatigue resistant properties. (Soundarapandiyan et al., 2023)

A. EXPERIMENTAL SETUP

The sample part was designed to exhibit four common type of anomalies and was performed on an EOS M290 equipped with Sigma Labs PrintRite3D system. The powder as specified above was a particle size distribution of 23-50 microns. The sample layout consisted of five separate regions. At far right, a control group was built. Adjoining to it at the left side were two parts each of embedded keyhole & lack of fusion spherical voids ranging from 1.00 mm – 0.05 mm diameter spheres. Followed by a set of parts with gas porosity induced due to anhydrous sodium acetate at an amount of 1.0 gm, 0.5 gm & 0.25gm per layer. Lastly, the parts were contaminated with tungsten powder with a particle distribution of 20-50 microns at the same layer height as the previous one.

B. COMPUTED TOMOGRAPHY ANALYSIS

CT was considered as the base for generating training labels to be used in supervised machine learning. The PrintRite3D in-process metrics were examined and one part for each anomaly type based on metric activity was chosen.(Beckett, n.d.)

C. REGISTRATION OF CT RESULTS

This advanced imaging technology was possible due to synchrotron radiation micro-computed tomography. Where, synchrotron radiation is produced by charged particles, typically electrons, moving at nearly speed of light in a circular or elliptical path within a synchrotron facility. μ-CT is a non-destructive imaging technique that uses X-rays for creating cross-sectional images of an object, allowing for 3D reconstruction. (Otto et al., 2022)

The anomalies appear as tessellations around the detected voids and inclusions. Since the coordinate system in these files is same as the coordinate system used in the STL files specifying the build, the process for registering them is simple. Then the ranges of x, y and z coordinates in the STL files were extracted and matched against x, y and z layer coordinates in the TED metrics.

The tessellations spanned for several layers, so they were projected onto individual layers as polygons. Theses polygons were allotted weighted sums considered as the length of enclosing polygons, to emphasize weight of larger voids. The weights were observed to be large positive integers which were normalized with the function: -

$\frac{2}{e^{-x}+1} - 1$, where x is label value

D. PRINTRITE ML MODEL (RANDOM FOREST)

The quality metrics, namely, Transformative Embedding Distillation, TED Sigma, Textual Entailment Prediction & TEP Sigma were added with internal metrics for calculating TEP Sigma, namely, the counts and the top sums. During training of models, the inclusion of these features & other potential candidates by checking the feature importance were considered for evaluation. After discarding regression models, the classification models were decided upon which needed ensemble methods such as forests, bagging or boosting. Random Forest, one of the ensemble methods proved to give a great performance in terms of accuracy and speed.

Random Forests are a combination of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest.(*Randomforest2001*, n.d.)

The ensemble of trees selected are controlled by random state parameter, which then propagates new random state to each component estimator. These component estimators are decision trees whose prediction at each point is computed as the average of the prediction of individual trees at that point. To ameliorate over-training, 10-fold cross validation on the data was used. This meant that the data set was randomly partitioned into 10 equal folds, and for each test fold we train on the other 9 folds and evaluate on the selected test fold. Next a pair of cost functions to measure residuals in the positive and negative directions was developed.

OBSERVATIONS

In the case-study, the labels used are probabilities and hence prediction probabilities are chosen, while calculating residuals as floating-point differences. Accordingly, the normal data is sub-sampled and the anomalies are over-sampled when anomalies are in small numbers with replacement to bring the counts into approximate balance and then calculate class weights to account for any remainder.

It was decided to diagnose the comparison between model prediction images with trained labels using a heat map. (Beckett, n.d.)

RESULTS

ADVERSARIAL MODELLING

Adversarial modelling helps in improving the robustness and security of machine learning systems. It is the practice of creating models to poke and prod existing models for finding weaknesses or vulnerabilities. These adversarial models throw curveballs trying to confuse existing models into making mistakes. In this case-study, models are trained to recognize their own anomaly types and avoid flagging other anomaly types. Adversarial modelling is used to handle rejection of the 'other' anomaly types, which can be enhanced by explicitly training against them.

Lack of Fusion Model

The layer cost values from the binary classification of LOF model, testing the parts with 4 anomaly types is showcased in the table below. The model trained with lack of fusion data rejects other anomalies while considering them false positives as implied from the low value on the same type and high value on the other types. Adjoining figures show the test layer visualization with respect to keyhole, gas porosity & tungsten anomalies.

LOF model	Total Cost	Positive Cost	Negative Cost
Lack of Fusion	1186	68	118
Keyhole	2105	108	1997
Gas Porosity	3604	141	3463
Tungsten	24192	24	24169

Another evaluation model is the visualization of predictions in the PrintRite3D user interface. In this case, the anomalies show up high probabilities, whereas keyhole, gas porosity and tungsten anomalies have low probabilities of being lack of fusion anomalies. The figure below shows the prediction on PrintRite3D visualization generated by the LOF model. This 3D tool ensures LOF model recognition parameters on parts apart from the trained part while ignoring most of the other anomaly types on the build plate. Similarly, the case-study consists of training model for other anomaly defects including Gas Porosity model, Keyhole model and Tungsten model.

CONCLUSION

Additive Manufacturing, a production technology though perceived as a mean to ease the manufacturing lead time and an innovative approach of producing functional products as compared to the current conventional industry techniques – the intended realisation of the technology is not yet completely actualized. One of the few factors for mustering those fulfilments is the attainment of arduous Quality Assurance parameters, which this case-study can be a rough manual for implementing Machine Learning aspects to the technology. It is clearly showcased in this case-study that four different types of anomalies: lack of fusion, gas porosity, keyhole and tungsten inclusion can be recognized by training ML models. These models were using PrintRite3D metrics including TED, TED Sigma, TEP and TEP Sigma as features and the CT data as labels. This work done in the case-study was focused on producing IPQM for each anomaly type. Prediction metrics like these can make it possible in the future to close the loop on controlling the build process. When production managers observe that anomalies are predicted, they can decide whether to interrupt the build process.

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